THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 1: Relations and Functions (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	Let R be the relation on the set $\{1,2,3\}$ given by $R = \{(1,1),(2,2),(3,3),(1,2),(2,3)\}$. Then R is: (A) reflexive, but neither symmetric nor transitive (B) an equivalence relation (C) symmetric and transitive only (D) transitive only	1
2.	The number of equivalence relations on the set $\{1,2,3\}$ containing $(1,2)$ and $(2,1)$ is: (A) 1 (B) 2 (C) 3 (D) 4	1
3.	The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is: (A) one-one and onto (B) many-one and onto (C) one-one but not onto (D) neither one-one nor onto	1
4.	The function $f: \mathbb{N} \to \mathbb{N}$ defined by $f(x) = 2x$ is: (A) bijective (B) injective but not surjective (C) surjective but not injective (D) neither injective nor surjective	1

Q No.	Question	Marks
5.	If set A has 3 elements and set B has 4 elements, then the number of one-one functions from A to B is: (A) 12 (B) 24 (C) 64 (D) 81	1
6.	**Assertion (A):** The relation $R = \{(a, b): a \leq b\}$ on the set of real numbers is not an equivalence relation. **Reason (R):** The relation $R = \{(a, b): a \leq b\}$ is not symmetric. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Show that the function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3 - 4x$ is a bijection.	2
	LEVEL 2 **Q 8 – Q 14**	**19**
8.	Show that the relation R on the set of integers $\mathbb{Z}$ defined by $R = \{(a, b): 2 \text{ divides } (a - b)\}$ is an equivalence relation.	2
9.	Check whether the relation R on $\mathbb{R}$ defined by $R = \{(a, b): a \leq b^3\}$ is reflexive, symmetric or transitive.	3
10.	Let $A = \{1,2,3, \ldots, 9\}$ and R be the relation on $A \times A$ defined by $(a, b)\ R\ (c, d)$ if $a + d = b + c$. Prove that R is an equivalence relation and find the equivalence class $[(2,5)]$.	3
11.	Let $f: \mathbb{N} \to \mathbb{N}$ be defined by $f(n) = \begin{cases} \frac{n+1}{2}, & n \text{ odd} \\ \frac{n}{2}, & n \text{ even} \end{cases}$. Show that f is onto but not one-one.	3
12.	Show that the function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x}{x^2+1}$ is neither one-one nor onto.	3
13.	Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Show that $f: A \to B$ defined by $f(x) = \frac{x-2}{x-3}$ is one-one and onto.	3
14.	Let $A = \{1,2,3,4,5,6\}$ and $R = \{(a, b): b = a + 1\}$ on A. Is R reflexive, symmetric or transitive? Justify.	2
	LEVEL 3 **Q 15 – Q 20**	**25**
15.	Let L be the set of all lines in a plane and $R = \{(L_1, L_2): L_1 \perp L_2\}$. Show that R is symmetric but neither reflexive nor transitive.	3
16.	Let $\mathbb{N}$ be the set of natural numbers and R be the relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\ R\ (c, d)$ if and only if $ad = bc$. Show that R is an equivalence relation.	5
17.	Show that the function $f: \mathbb{R} \to \{x \in \mathbb{R}: -1 < x < 1\}$ defined by $f(x) = \frac{x}{1+\lvert x \rvert}$ is one-one and onto.	5

Q No.	Question	Marks
18.	Let $A = \{x \in \mathbb{Z}: 0 \leq x \leq 12\}$. Show that $R = \{(a, b): \lvert a - b \rvert$ is a multiple of $4\}$ is an equivalence relation on A. Find the set of all elements related to 1 and write the equivalence class [2].	5
19.	Show that the function $f: [-1,1] \to \mathbb{R}$ defined by $f(x) = \frac{x}{x+2}$ is one-one. Also find the range of f.	3
20.	**Case Study:** A classroom teacher wants to assess her students' understanding of "relations". She writes the following five relations, each defined on the set $A = \{1,2,3\}$: $R_1 = \{(2,3), (3,2)\}$, $R_2 = \{(1,2), (1,3), (3,2)\}$, $R_3 = \{(1,2), (2,1), (1,1)\}$, $R_4 = \{(1,1), (1,2), (3,3), (2,2)\}$, $R_5 = \{(1,1), (1,2), (3,3), (2,2), (2,1), (2,3), (3,2)\}$. **(i)** Identify the relation which is reflexive, transitive but not symmetric. **[1]** **(ii)** Identify the relation which is reflexive and symmetric but not transitive. **[1]** **(iii)** (a) Identify the relations which are symmetric but neither reflexive nor transitive. **OR** (b) What pairs should be added to the relation R_2 to make it an equivalence relation? **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 1: Relations and Functions

Q No.	Answer / Key Steps
	LEVEL 1
1.	(A) Reflexive; not symmetric since $(1,2) \in R$ but $(2,1) \notin R$; not transitive since $(1,2), (2,3) \in R$ but $(1,3) \notin R$.
2.	(B) 2
3.	(D) $f(-1) = f(1)$ so many-one; negative numbers have no pre-image so not onto.
4.	(B) injective but not surjective (odd numbers have no pre-image).
5.	(B) $4 \times 3 \times 2 = 24$
6.	(A) Both true and R explains A.
7.	$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ (one-one). For any y, $x = \frac{3-y}{4}$ gives $f(x) = y$ (onto). Hence bijective.
	LEVEL 2
8.	Reflexive: $a - a = 0$ divisible by 2. Symmetric: $2 \mid (a-b) \Rightarrow 2 \mid (b-a)$. Transitive: $(a-b) + (b-c) = a - c$ divisible by 2. Hence equivalence.
9.	Not reflexive: $\frac{1}{2} \not\leq \frac{1}{8}$. Not symmetric: $(1,2) \in R$ but $(2,1) \notin R$. Not transitive: $\left(3, \frac{3}{2}\right), \left(\frac{3}{2}, \frac{4}{3}\right) \in R$ but $3 \not\leq \left(\frac{4}{3}\right)^3 = \frac{64}{27}$.
10.	$(a,b)R(c,d) \Leftrightarrow a - b = c - d$; reflexive, symmetric, transitive follow. $[(2,5)] = \{(1,4), (2,5), (3,6), (4,7), (5,8), (6,9)\}$.
11.	$f(1) = f(2) = 1$, so not one-one. For any $m \in \mathbb{N}$, $f(2m) = m$, so onto.
12.	$f(2) = f\left(\frac{1}{2}\right) = \frac{2}{5}$, so not one-one. $\lvert f(x) \rvert \leq \frac{1}{2}$ for all x, so e.g. 1 has no pre-image; not onto.
13.	One-one: $\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3} \Rightarrow x_1 = x_2$. Onto: for $y \neq 1$, $x = \frac{3y-2}{y-1} \in A$ and $f(x) = y$.
14.	Not reflexive ($(1,1) \notin R$), not symmetric ($(1,2) \in R, (2,1) \notin R$), not transitive ($(1,2), (2,3) \in R, (1,3) \notin R$).
	LEVEL 3
15.	A line is not perpendicular to itself (not reflexive). $L_1 \perp L_2 \Rightarrow L_2 \perp L_1$ (symmetric). $L_1 \perp L_2, L_2 \perp L_3 \Rightarrow L_1 \parallel L_3$ (not transitive).
16.	Reflexive: $ab = ba$. Symmetric: $ad = bc \Rightarrow cb = da$. Transitive: $ad = bc$ and $cf = de$ give $adcf = bcde \Rightarrow af = be$. Hence equivalence.
17.	For $x \geq 0$, $f(x) = \frac{x}{1+x}$; for $x < 0$, $f(x) = \frac{x}{1-x}$; each piece is one-one and signs separate them, so f is one-one. For $0 \leq y < 1$ take $x = \frac{y}{1-y}$; for $-1 < y < 0$ take $x = \frac{y}{1+y}$. So onto.
18.	Equivalence (as $\lvert a-a \rvert = 0$, $\lvert a-b \rvert = \lvert b-a \rvert$, $a - c = (a-b) + (b-c)$). Elements related to 1: $\{1,5,9\}$; $[2] = \{2,6,10\}$.
19.	$f(x_1) = f(x_2) \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$. $f'(x) = \frac{2}{(x+2)^2} > 0$ so f increasing; range $= [f(-1), f(1)] = \left[-1, \frac{1}{3}\right]$.
20.	(i) R_4. (ii) R_5 (not transitive: $(1,2), (2,3) \in R_5$ but $(1,3) \notin R_5$). (iii)(a) R_1 and R_3. (iii)(b) Add $(1,1), (2,2), (3,3), (2,1), (3,1), (2,3)$, so that R_2 becomes $A \times A$.